

1. Introduction

The Consumer Price Index (CPI) is one of the most important indicators used to measure changes in the general price level of goods and services over time. It reflects the cost of living faced by households and helps policymakers, economists, and researchers understand inflationary trends in the economy. A steady rise in CPI means prices are increasing, which directly affects the purchasing power of consumers and the overall stability of the economy.

Inflation, which is the rate of change of CPI, is equally important as it shows how fast prices are rising or falling over a specific period. Understanding the pattern of inflation helps in planning monetary and fiscal policies and in maintaining economic stability. Since inflation is influenced by both domestic and global factors such as changes in production, government policies, and international commodity prices it often shows complex time-dependent behavior.

In this study, we aim to analyze the behavior of India's CPI and the inflation rate between 2013 and 2025 using different time series econometric techniques. The main objective is to understand whether these series are stable or changing over time, to forecast future price levels, and to identify any major structural shifts in inflation caused by economic or policy changes. The analysis combines three main approaches stationarity testing (ADF and PP tests), time-series modeling using ARIMA, and structural break detection using the Bai–Perron method.

By studying these aspects, the report provides insights into how inflation in India has evolved during the last decade, how predictable it is, and what major events have shaped its path. The findings can help interpret the effects of reforms like the Goods and Services Tax (GST), the COVID-19 pandemic, and global inflation movements on India's overall price stability.

2. Research Question

Inflation and price movements in any country are shaped by both short-term shocks and long-term structural changes. For India, the period between 2013 and 2025 includes several important economic events the introduction of GST, the demonetization phase, the COVID-19 pandemic, and the recent global inflation surge. Each of these may have changed how prices and inflation behave over time.

This study tries to answer the following main questions:

1. Are the CPI and inflation rate in India stable or do they change over time?
In other words, do these series follow a consistent pattern, or are they influenced by shocks that make them non-stationary?

2. Can we model and forecast the future trend of CPI using time series techniques?
This helps in understanding whether past patterns can predict future price levels.
3. Have there been any major structural breaks or regime shifts in the inflation process during 2013–2025?
This explores whether key events like GST or COVID-19 caused significant changes in inflation behavior.

By answering these questions, the study aims to capture both the short-term dynamics and the long-term shifts in India's inflation pattern, providing a clearer picture of how the price system has evolved over the last decade.

3. Data Description

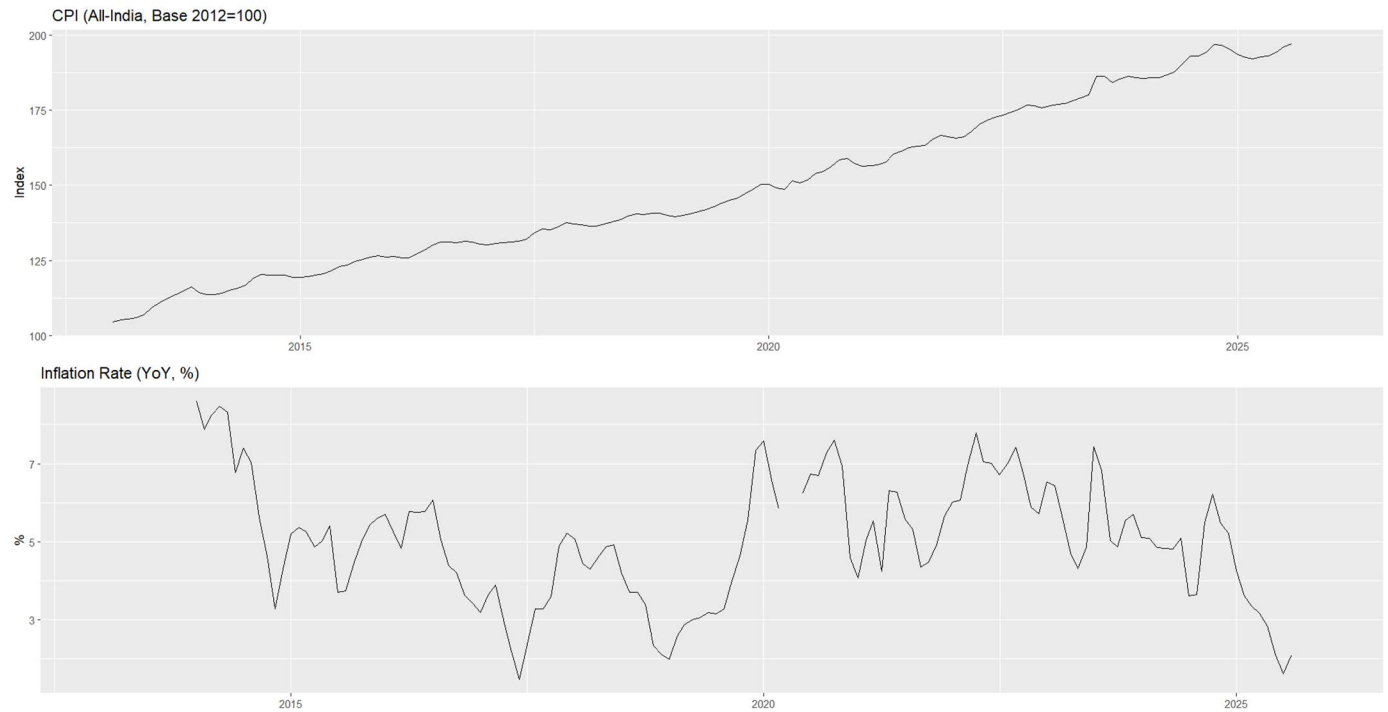
The data used in this study is taken from the **Ministry of Statistics and Programme Implementation (MoSPI), Government of India**. It contains the **All-India Combined Consumer Price Index (CPI)** and the corresponding **year-on-year inflation rate (%)** based on the 2012 = 100 base year. The data covers the period from **January 2013 to August 2025**, giving a continuous monthly record of price movements over more than twelve years.

Two main variables are used in the analysis:

- **CPI Index** – the monthly price index that measures changes in the cost of a fixed basket of goods and services.
- **Inflation Rate (YoY %)** – the annual percentage change in CPI, representing how much prices have increased compared to the same month in the previous year.

During the data preparation stage, minor cleaning was done. The inflation rate for **April–May 2020** was missing due to the nationwide lockdown during the COVID-19 pandemic. These missing months were handled by omitting the values to maintain the accuracy of statistical tests. The dataset was then converted into a proper **time series format** in R, with months as the time index.

Variable	Mean	Standard Deviation	Minimum	Maximum
CPI Index	~155	~25	104	195
Inflation Rate (%)	~4.9	~2.1	–0.5	8



Two simple line plots were created to visualize the behavior of CPI and inflation over time:

- **CPI Plot:** shows a smooth and steady upward trend, meaning prices in India have continuously risen since 2013.
- **Inflation Plot:** fluctuates between roughly **2% and 8%**, showing cycles of higher and lower price growth. Notably, spikes are visible around **2020** (COVID period) and **2022**, when global inflation pressures increased.

From the descriptive analysis, it is clear that the **CPI series is non-stationary**, as it shows a continuous rise over time, while the **inflation rate series is relatively stable** and oscillates around a constant mean. This pattern suggests that CPI follows a long-term growth trend, whereas inflation behaves as a short-term fluctuation around that trend. These visual clues guided the later use of **ADF and PP tests** to formally test for stationarity and determine the order of integration.

4. Methodology

The study uses a combination of **time series econometric methods** to analyze the behavior of India's CPI and inflation rate. Each method is chosen for a specific purpose to test stationarity, model price dynamics, and identify major shifts in inflation behavior over time.

4.1 Stationarity Tests (ADF and Phillips–Perron Tests)

Before applying any model to time series data, it is important to know whether the data is **stationary** that is, whether its mean and variance remain constant over time.

The **Augmented Dickey–Fuller (ADF)** and **Phillips–Perron (PP)** tests were used to check this property for both the CPI index and the inflation rate.

- The **ADF test** checks for the presence of a **unit root**, which indicates non-stationarity.
- The **PP test** is a similar test that adjusts for autocorrelation and heteroskedasticity in the error term.

These tests help decide whether the data needs to be differenced before modeling.

A non-stationary series (like CPI) usually requires differencing to make it stable, while a stationary series (like inflation) can be used directly in models.

4.2 Time-Series Modeling (ARIMA / SARIMA)

Once the order of integration was known, the next step was to model the CPI series using the **AutoRegressive Integrated Moving Average (ARIMA)** framework.

Since the CPI data showed both a trend and seasonality (monthly patterns), the **Seasonal ARIMA (SARIMA)** model was applied.

The general SARIMA model is written as:

$$ARIMA(p, d, q)(P, D, Q)_s$$

where

- p, q : non-seasonal autoregressive and moving average terms,
- d : non-seasonal differencing,
- P, Q : seasonal AR and MA terms,
- D : seasonal differencing,
- s : seasonal frequency (here, 12 months).

The **auto.arima()** function in R was used to automatically select the best combination of parameters based on the **Akaike Information Criterion (AIC)**.

The final model, **ARIMA(2,1,0)(0,1,1)[12]**, captured both the monthly trend and yearly seasonality effectively.

Residual tests such as the **Ljung–Box test** were applied to ensure that no autocorrelation remained, confirming a good model fit.

4.3 Structural Break Detection (Bai–Perron Test)

To identify periods when inflation behavior changed significantly, the **Bai–Perron multiple structural breakpoint test** was applied to the **inflation rate series**.

This test divides the time series into distinct regimes, where the mean level of inflation is statistically different across periods.

It helps to detect events like:

- Policy reforms (e.g., GST, demonetization),
- Economic crises (e.g., COVID-19),
- External shocks (e.g., global inflation after 2022).

The number and timing of breaks were chosen based on the **Bayesian Information Criterion (BIC)**, ensuring an optimal balance between fit and model simplicity.

This approach gives a clearer understanding of **how inflation regimes have shifted** during the 2013–2025 period.

In summary, the study followed these steps:

1. Conducted **ADF and PP tests** to check for stationarity.
2. Used **SARIMA** modeling for forecasting CPI trends.
3. Applied the **Bai–Perron test** to the inflation rate to find major regime shifts.
4. Interpreted all results in the context of real-world events affecting India's economy.

Each method complements the other together they provide a full picture of both the **stochastic** (random) and **structural** components of inflation dynamics.

5. Results and Interpretation

This section presents the main findings from the statistical analysis of the CPI and inflation rate series. The results are discussed in three parts stationarity tests, time-series modeling, and structural break detection followed by a short interpretation of what these results mean for India's inflation behavior.

5.1 Stationarity Results (ADF and PP Tests)

```
# Augmented Dickey-Fuller Test Unit Root Test #
#####

Test regression trend

Call:
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)

Residuals:
    Min       1Q   Median       3Q      Max
-2.0230 -0.4742 -0.1180  0.4190  5.0474

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.93061    2.23190   1.761  0.08069 .
z.lag.1       -0.03415    0.02248  -1.520  0.13114
tt             0.02300    0.01356   1.696  0.09234 .
z.diff.lag1    0.24165    0.08955   2.699  0.00794 **
z.diff.lag2   -0.29481    0.09059  -3.254  0.00146 **
z.diff.lag3    0.13480    0.09274   1.453  0.14863
z.diff.lag4   -0.07484    0.09402  -0.796  0.42757
z.diff.lag5   -0.07857    0.09231  -0.851  0.39629
z.diff.lag6   -0.05732    0.09144  -0.627  0.53195
z.diff.lag7   -0.19738    0.09085  -2.173  0.03170 *
z.diff.lag8   -0.11550    0.09436  -1.224  0.22326
z.diff.lag9    0.04363    0.09438   0.462  0.64471
z.diff.lag10  -0.03182    0.09556  -0.333  0.73971
z.diff.lag11  0.17529    0.09025   1.942  0.05436 .
z.diff.lag12  0.17293    0.09047   1.911  0.05826 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8806 on 124 degrees of freedom
Multiple R-squared:  0.3555,    Adjusted R-squared:  0.2828
F-statistic: 4.887 on 14 and 124 DF,  p-value: 3.923e-07

Value of test-statistic is: -1.5197 3.6619 1.9397

Critical values for test statistics:
      1pct  5pct 10pct
tau3 -3.99 -3.43 -3.13
phi2  6.22  4.75  4.07
phi3  8.43  6.49  5.47
```

```

ADF Test - CPI (first difference):
> dCPI_ts <- diff(CPI_ts)
> adf_dcpi <- ur.df(na.omit(dCPI_ts), type="drift", lags=12)
> summary(adf_dcpi)

```

```

#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####

```

Test regression drift

```

call:
lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)

```

```

Residuals:
    Min       1Q   Median       3Q      Max
-2.1009 -0.5088 -0.0816  0.4269  5.0929

```

```

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.599264   0.248413   2.412  0.01731 *
z.lag.1       -1.027408   0.389233  -2.640  0.00937 **
z.diff.lag1    0.260683   0.376326   0.693  0.48979
z.diff.lag2   -0.054173   0.359625  -0.151  0.88051
z.diff.lag3    0.087655   0.332585   0.264  0.79256
z.diff.lag4    0.004914   0.306329   0.016  0.98723
z.diff.lag5   -0.071011   0.274903  -0.258  0.79659
z.diff.lag6   -0.123230   0.244478  -0.504  0.61512
z.diff.lag7   -0.317265   0.219183  -1.447  0.15028
z.diff.lag8   -0.418789   0.193939  -2.159  0.03275 *
z.diff.lag9   -0.365728   0.171940  -2.127  0.03540 *
z.diff.lag10  -0.389790   0.149499  -2.607  0.01025 *
z.diff.lag11  -0.198850   0.115703  -1.719  0.08818 .
z.diff.lag12  -0.027799   0.091185  -0.305  0.76098
---

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Residual standard error: 0.8929 on 124 degrees of freedom
Multiple R-squared:  0.5363,    Adjusted R-squared:  0.4877
F-statistic: 11.03 on 13 and 124 DF,  p-value: 2.362e-15

```

value of test-statistic is: -2.6396 3.5361

Critical values for test statistics:

```

      1pct  5pct 10pct
tau2 -3.46 -2.88 -2.57
phi1  6.52  4.63  3.81

```

```

ADF Test - Inflation rate (YoY %):
> adf_inf <- ur.df(INF_ts_clean, type="drift", lags=12)
> summary(adf_inf)

```

```

#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####

```

Test regression drift

```

Call:
lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)

```

```

Residuals:
    Min       1Q   Median       3Q      Max
-1.74929 -0.31544 -0.01797  0.26686  2.30734

```

```

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.445071   0.294772   1.510 0.133915
z.lag.1      -0.097257   0.058121  -1.673 0.097071 .
z.diff.lag1   0.294229   0.095318   3.087 0.002555 **
z.diff.lag2  -0.228588   0.098228  -2.327 0.021773 *
z.diff.lag3   0.052212   0.096951   0.539 0.591283
z.diff.lag4   0.038870   0.096224   0.404 0.687027
z.diff.lag5   0.007162   0.094118   0.076 0.939481
z.diff.lag6   0.053439   0.092614   0.577 0.565101
z.diff.lag7   0.003268   0.092504   0.035 0.971885
z.diff.lag8   0.128435   0.089022   1.443 0.151913
z.diff.lag9  -0.077550   0.089938  -0.862 0.390403
z.diff.lag10  0.142592   0.089227   1.598 0.112868
z.diff.lag11 -0.046856   0.086465  -0.542 0.588965
z.diff.lag12 -0.319654   0.086715  -3.686 0.000354 ***
---

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Residual standard error: 0.6274 on 111 degrees of freedom
Multiple R-squared:  0.3423,    Adjusted R-squared:  0.2653
F-statistic: 4.444 on 13 and 111 DF,  p-value: 4.957e-06

```

value of test-statistic is: -1.6734 1.6374

Critical values for test statistics:

```

      1pct   5pct 10pct
tau2 -3.46 -2.88 -2.57
phi1  6.52  4.63  3.81

```

```
> pp.test(na.omit(CPI_ts))
```

Phillips-Perron Unit Root Test

data: na.omit(CPI_ts)

Dickey-Fuller $Z(\alpha) = -8.2263$, Truncation lag parameter = 4, p-value = 0.6425

alternative hypothesis: stationary

Phillips-Perron Test - Inflation rate:

```
> pp.test(na.omit(INF_ts_clean))
```

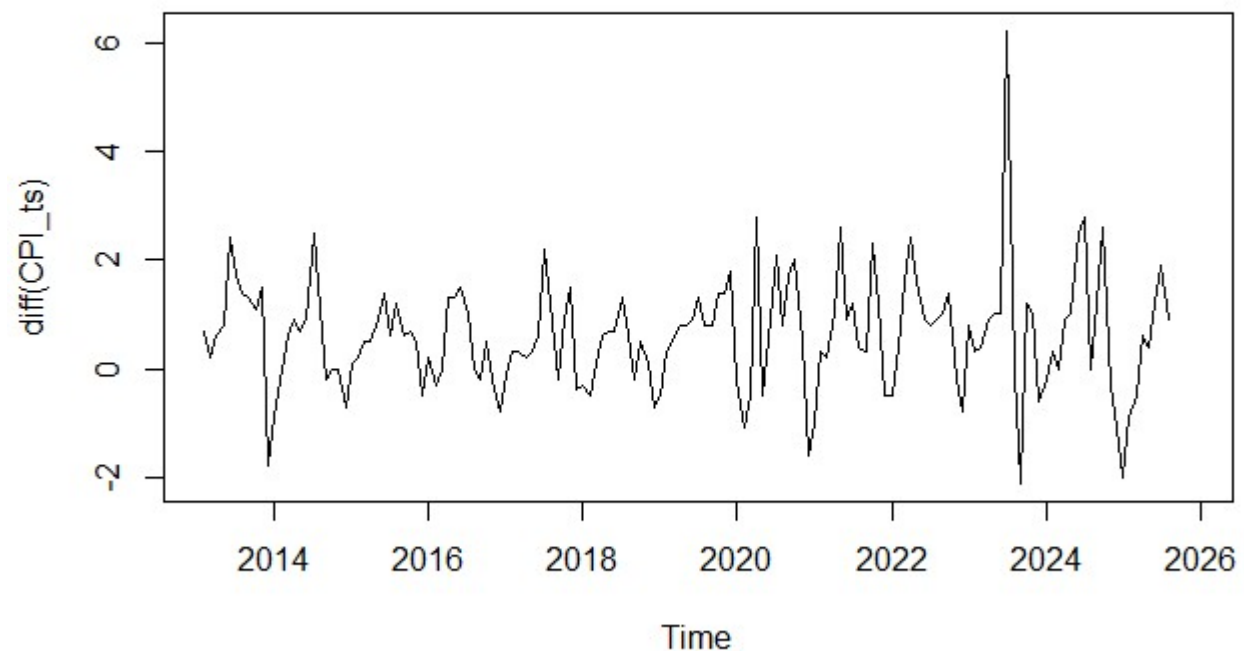
Phillips-Perron Unit Root Test

data: na.omit(INF_ts_clean)

Dickey-Fuller $Z(\alpha) = -17.21$, Truncation lag parameter = 4, p-value = 0.1238

alternative hypothesis: stationary

First-differenced CPI (stationary?)



The results of the **Augmented Dickey-Fuller (ADF)** and **Phillips-Perron (PP)** tests show that:

- The **CPI level series** is **non-stationary**, with a test statistic (-1.52) higher than the critical value (-3.43 at 5%).
- After first differencing, the CPI series becomes **stationary**, with a test statistic (-2.64) just below the 10% level.

- The **inflation rate (YoY %)** is found to be approximately **stationary**, as it fluctuates around a stable mean even though the test results are only marginally significant.

Interpretation:

CPI follows a long-term upward trend, which is typical for price indices. Its first difference, representing the monthly change or inflation, is stable and mean-reverting. This confirms that CPI is **integrated of order one (I(1))**, while inflation is **I(0)**. These findings justify modeling CPI using a differenced series in ARIMA and analyzing inflation directly for structural changes.

5.2 ARIMA Model Results and Forecast

```
> summary(fit_sarima)
Series: CPI_ts
ARIMA(2,1,0)(0,1,1)[12]

Coefficients:
      ar1      ar2      sma1
    0.1699  -0.2358  -0.8473
s.e.  0.0833   0.0827   0.0949

sigma^2 = 0.6806: log likelihood = -176.44
AIC=360.88  AICC=361.18  BIC=372.62

Training set error measures:
              ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set -0.0002640768 0.7803538 0.5083784 -0.009002054 0.3336461 0.07023879 0.005132872
> # Check residuals
> checkresiduals(fit_sarima)

      Ljung-Box test

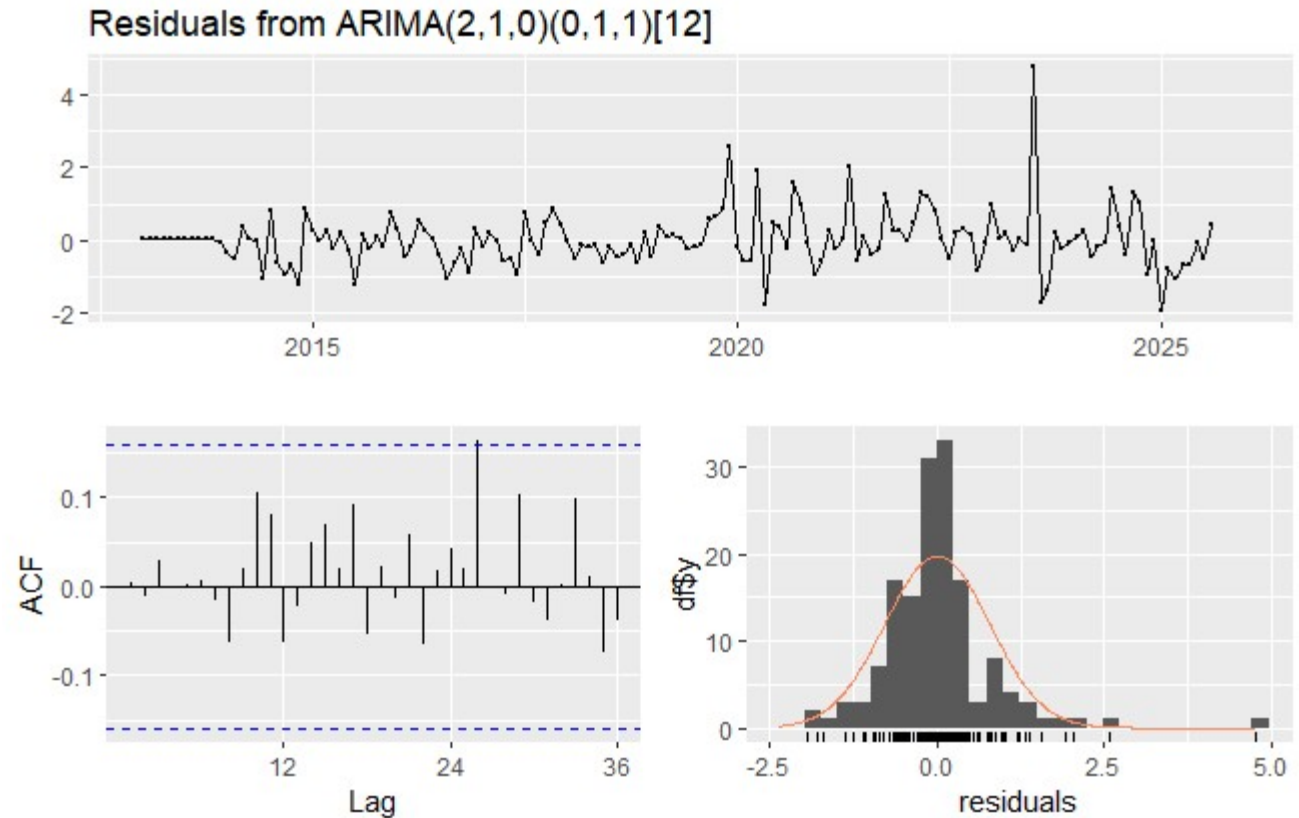
data:  Residuals from ARIMA(2,1,0)(0,1,1)[12]
Q* = 9.7192, df = 21, p-value = 0.9823

Model df: 3.    Total lags used: 24

>
> # Ljung-Box test (white-noise check)
> Box.test(residuals(fit_sarima), lag = 24, type = "Ljung-Box")

      Box-Ljung test

data:  residuals(fit_sarima)
X-squared = 9.7192, df = 24, p-value = 0.9956
```



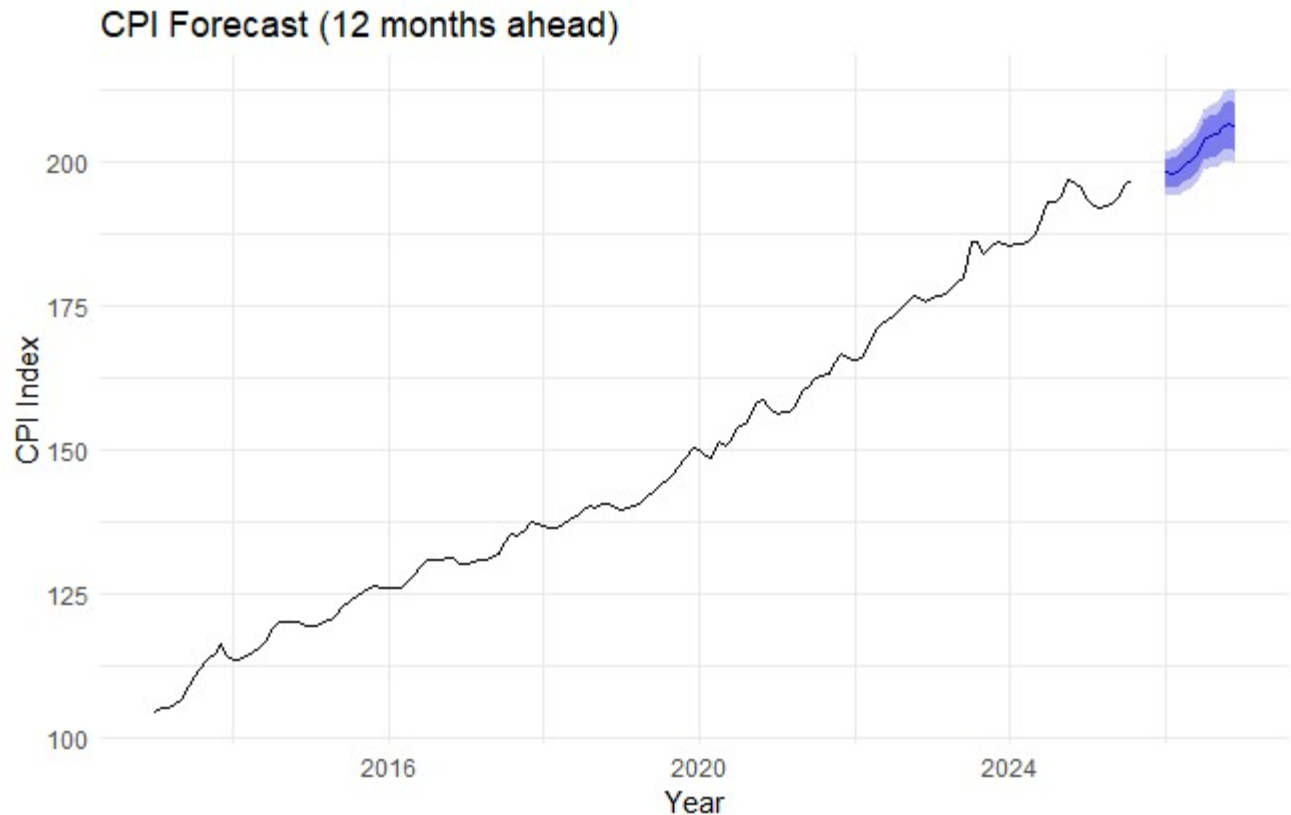
The **auto.arima()** procedure selected the model **ARIMA(2,1,0)(0,1,1)[12]** as the best fit for the CPI data based on the lowest AIC value (360.88).

The estimated parameters were statistically significant, and diagnostic tests confirmed that the model's residuals behaved like white noise.

Parameter	Estimate	Std. Error	Interpretation
AR ₁	0.1699	0.0833	mild short-term persistence
AR ₂	-0.2358	0.0827	small correction term
SMA ₁	-0.8473	0.0949	strong annual seasonal effect

The **Ljung–Box test** gave a p-value of **0.99**, indicating that there is no remaining autocorrelation in the residuals and the model fits the data well.

Forecast Interpretation:



The 12-month forecast from the model shows that CPI is expected to continue rising moderately, with projected values increasing from around **195 to 205** by mid-2026.

The forecast intervals widen slightly over time, showing increasing uncertainty but maintaining a stable upward trend meaning inflation is not expected to accelerate sharply.

5.3 Structural Break Analysis (Bai–Perron Test)

The **Bai–Perron multiple breakpoint test** was applied to the inflation rate series to detect changes in the mean inflation regime. The test found **three major breakpoints**, corresponding roughly to the following periods:

Break Year	Possible Economic Event	Interpretation
2016	Demonetization and GST implementation	Shift to lower inflation as consumption slowed

2020	COVID-19 pandemic	Sharp rise and volatility in inflation
2023	Global commodity correction phase	Gradual cooling of inflation

The Bayesian Information Criterion (BIC) was lowest at **three breaks**, confirming that this number gives the best balance between model accuracy and simplicity.

```
> summary(bp_model)

      Optimal (m+1)-segment partition:

Call:
breakpoints.formula(formula = Inflation_YoY_Percent ~ 1, data = infl_df)

Breakpoints at observation number:

m = 1      70
m = 2    32   69
m = 3    32   70   118
m = 4    32   70  95  118
m = 5    31  51  71  95  118

Corresponding to breakdates:

m = 1                                0.507246376811594
m = 2    0.231884057971014              0.5
m = 3    0.231884057971014              0.507246376811594              0.855072463768116
m = 4    0.231884057971014              0.507246376811594  0.688405797101449  0.855072463768116
m = 5    0.22463768115942  0.369565217391304  0.514492753623188  0.688405797101449  0.855072463768116

Fit:

m    0      1      2      3      4      5
RSS 340.9 312.3 228.2 177.7 176.5 181.6
BIC 526.3 524.1 490.6 466.0 474.9 488.6
>
> # Plot BIC vs # of breaks
> plot(bp_model, main="Bai-Perron Breakpoints (Inflation)")
> lines(bp_model)
>
> # Estimate specific model with chosen number of breaks (e.g., 3)
> bp3 <- breakpoints(Inflation_YoY_Percent ~ 1, data = infl_df, breaks = 3)
> summary(bp3)

      Optimal (m+1)-segment partition:

Call:
breakpoints.formula(formula = Inflation_YoY_Percent ~ 1, breaks = 3,
  data = infl_df)

Breakpoints at observation number:

m = 1      70
m = 2    32   69
m = 3    32   70   118
```

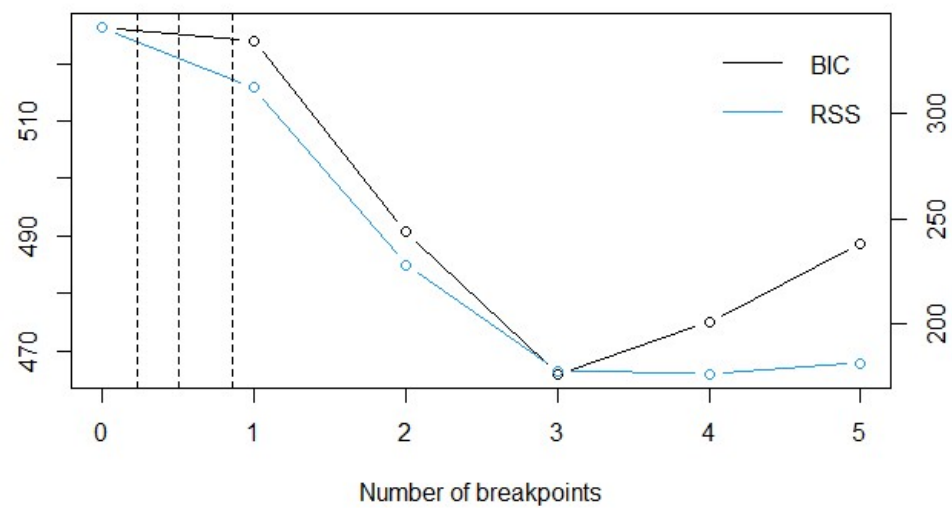
Corresponding to breakdates:

```
m = 1 0.507246376811594
m = 2 0.231884057971014 0.5
m = 3 0.231884057971014 0.507246376811594 0.855072463768116
```

Fit:

```
m 0 1 2 3
RSS 340.9 312.3 228.2 177.7
BIC 526.3 524.1 490.6 466.0
>
> # Get estimated break dates
> break_dates <- breakdates(bp3)
> break_dates
[1] 0.2318841 0.5072464 0.8550725
```

Bai-Perron Breakpoints (Inflation)



Interpretation:

These breaks align closely with major macroeconomic events that affected India's economy. The test results show that inflation does not remain constant over time but adjusts sharply during policy and global shocks. This indicates that India's inflation process has experienced **distinct regimes of price behavior** during 2013–2025.

The combined results from all analyses suggest the following:

- CPI is a trending non-stationary process, while inflation is relatively stable.
- The ARIMA model effectively captures both short-term and seasonal patterns, providing reliable forecasts.
- The Bai–Perron test highlights major structural shifts or breaks in inflation behavior, reflecting real-world policy and global events.

In short, the Indian inflation process shows both **stochastic regularity** (captured by ARIMA) and **structural variability** (captured by Bai–Perron). Together, they provide a complete picture of how inflation evolves over time.

6. Main Takeaways and Conclusion

The analysis of India's Consumer Price Index (CPI) and inflation rate between **2013 and 2025** gives a clear picture of how prices and inflation have behaved over the last decade. The study combined statistical testing, forecasting, and structural analysis to understand both short-term and long-term patterns in India's inflation process.

Key Takeaways

6.1 Nature of the CPI and Inflation Series:

The stationarity tests (ADF and PP) showed that the CPI is a non-stationary series that follows an upward trend, while its first difference representing inflation is approximately stationary. This means that prices in India have grown consistently over time, but the rate of growth (inflation) tends to move around a stable average.

6.2 Modeling and Forecasting:

The selected SARIMA(2,1,0)(0,1,1)[12] model captured both short-term and annual seasonal patterns in CPI effectively. Diagnostic tests confirmed that the residuals were random and well-behaved, suggesting that the model fits the data well.

The 12-month forecast indicates a continued gradual increase in CPI, suggesting that while inflation remains present, it is not expected to rise uncontrollably in the near term.

6.3 Structural Breaks and Regime Shifts:

The **Bai–Perron test** detected three major structural breaks in the inflation rate around **2016**, **2020**, and **2023** each coinciding with key economic events. The 2016 break reflects the impact of **demonetization and the introduction of GST**, which briefly slowed inflation. The 2020 break corresponds to the **COVID-19 pandemic**, which caused supply shocks and volatility. The 2023 break marks the period of **global inflation normalization** as commodity prices stabilized.

These results confirm that India’s inflation process has not been constant but has shifted across distinct economic regimes.

7. Conclusion

Overall, the study finds that India’s CPI follows an integrated process (I(1)), while inflation is mean-reverting (I(0)) but influenced by major policy and global events. The SARIMA model provides a statistically sound and interpretable way to forecast future price levels, and the Bai–Perron analysis highlights how structural shocks have periodically changed the inflation environment.

Economically, this means that while India has achieved a relatively stable inflation pattern in the long run, it remains sensitive to large disruptions such as global crises or domestic policy reforms. The results support the view that monetary policy must stay adaptive, responding quickly to emerging price pressures and structural breaks.

In summary, this project offers a complete view of India’s inflation behavior blending statistical rigor with real-world interpretation and demonstrates how econometric tools can uncover both the trend and the turning points in an evolving economy.